

CompTIA SecAI+

Setup Guide

Getting a Groq API Key and Connecting it to Open WebUI

From creating a free account to chatting with GPT OSS 20B

Before you start

This guide walks you through setting up Groq to run the SecAI+ practicals. Groq is a fast, free AI service that connects to Open WebUI the same way Gemini and OpenRouter did. You will create your own free Groq account, generate an API key, paste it into Open WebUI, and select the model to use.

There are four short parts:

- Part A — Create a Groq account and get your API key.
- Part B — Turn on Direct Connections in Open WebUI (one-time).
- Part C — Add your Groq key to Open WebUI.
- Part D — Select the model (GPT OSS 20B) and start.

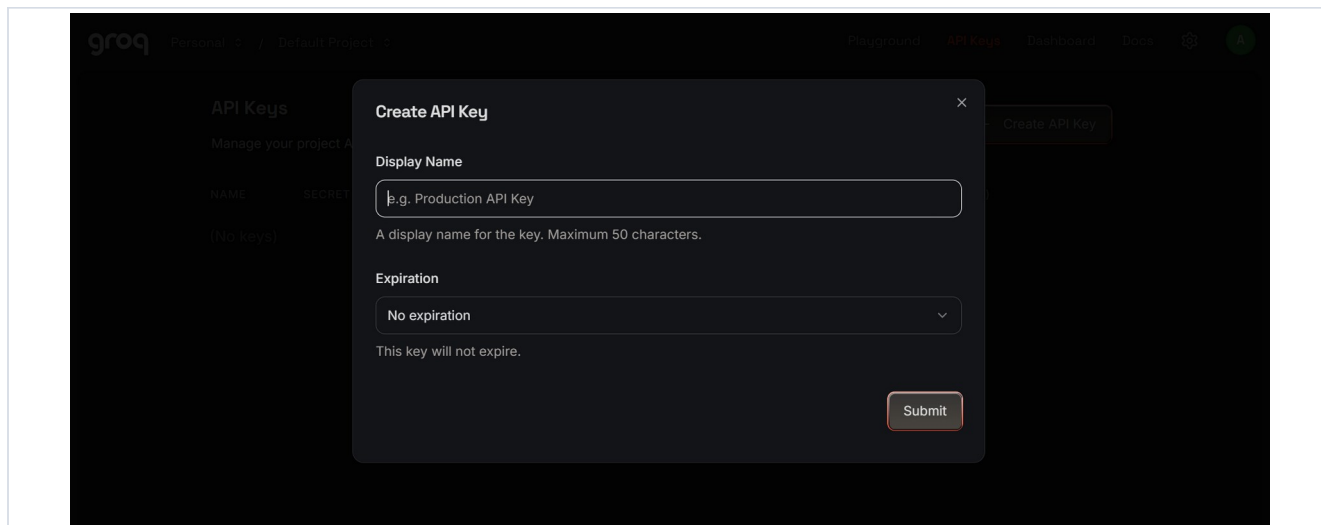
Which model? For all the practicals, use GPT OSS 20B. It is fast and light on the free limits. Keep GPT OSS 120B as a backup if you ever want more detailed answers.

Everyone needs their own key. Each participant should make their own Groq account and key. The free limits are counted per account, so sharing one key means sharing one small daily allowance.

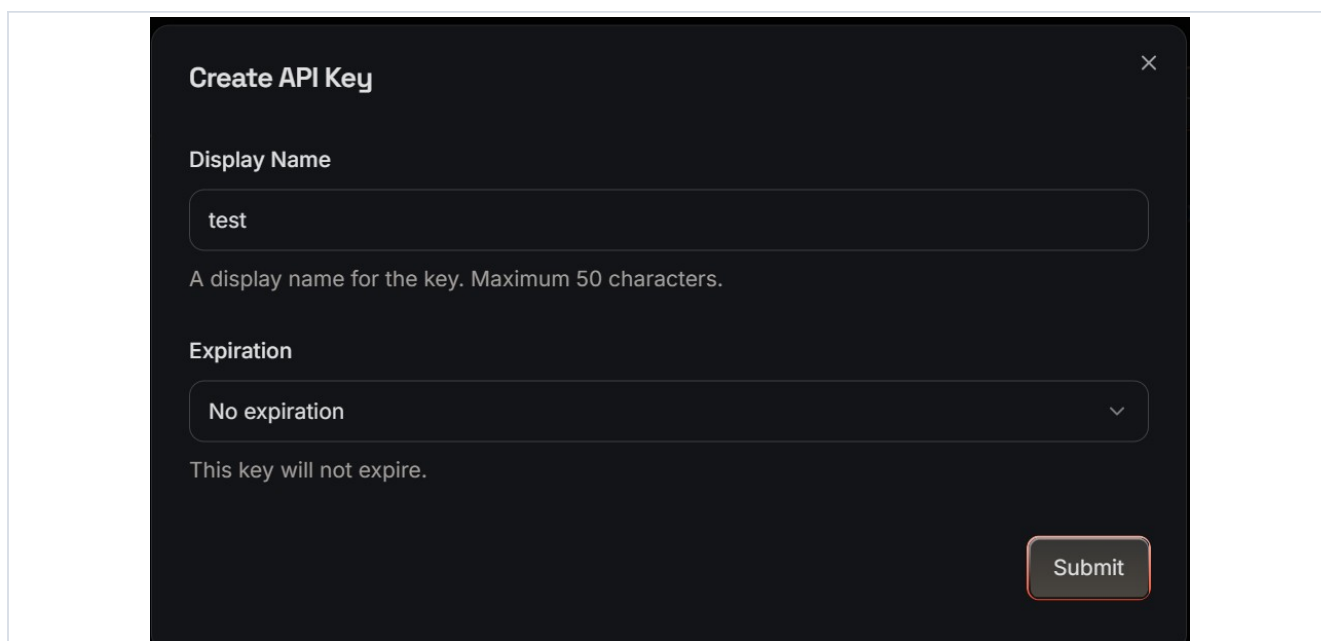
Part A — Create a Groq account and get your API key

Step 1 Open your browser and go to **console.groq.com**. Click **Log In**, then sign in the easy way with **Continue with Google** or **Continue with GitHub** (you can also use email). No credit card is needed.

Step 2 Once you are signed in, click **API Keys** in the top menu, then click the **Create API Key** button on the right. A small box will open.



Step 3 In the **Display Name** field, type a simple name like **test**. Leave **Expiration** set to **No expiration**, then click **Submit**.



Step 4 Your key is now shown **once**. Click **Copy** and paste it somewhere safe (a notepad file). The key starts with **gsk_**. You will paste it into Open WebUI in Part C.

Create API Key



Your new API key has been created. **Copy it now, as we will not display it again.**

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

 Copy

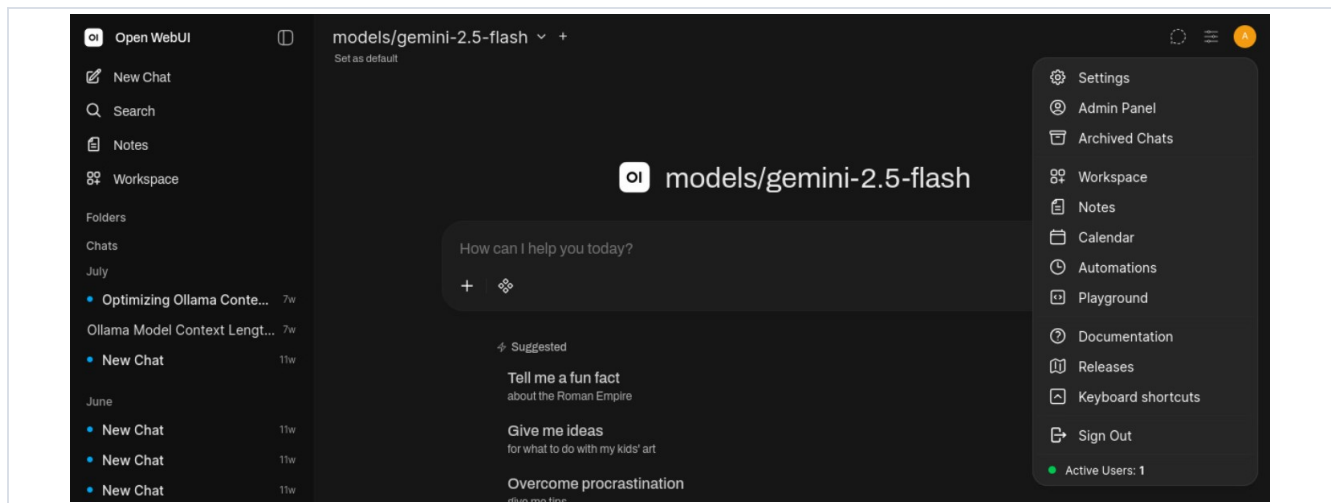
Done

Important Groq will not show this key again after you close the box. If you lose it, just create another key — it is free.

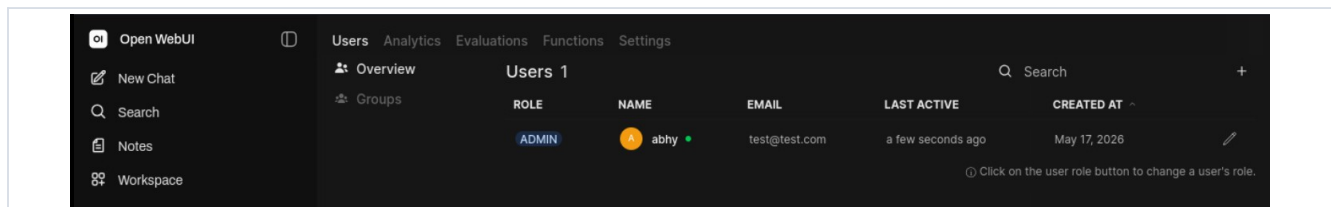
Part B — Turn on Direct Connections in Open WebUI

This is a one-time switch that lets Open WebUI accept your own API keys.

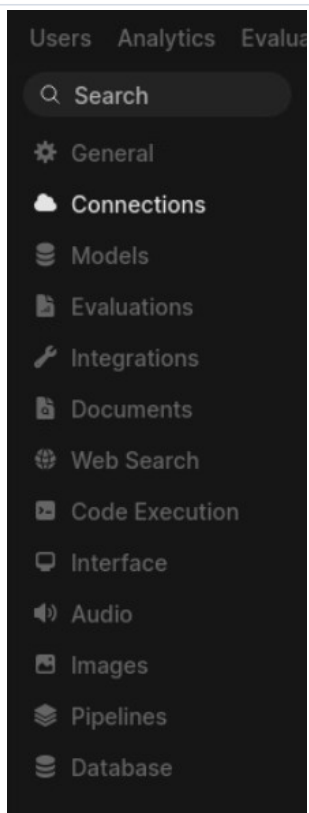
Step 5 In Open WebUI, click your **profile icon** (the coloured circle, top-right), then click **Admin Panel**.



Step 6 In the Admin Panel, click the **Settings** tab at the top.

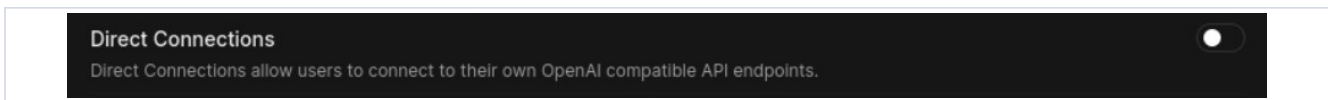


Step 7 From the left menu, click **Connections**.

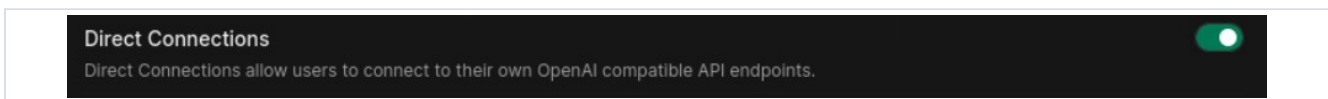


Step 8 Find the **Direct Connections** switch and turn it **ON** (it turns green). Click **Save**.

Before (off):

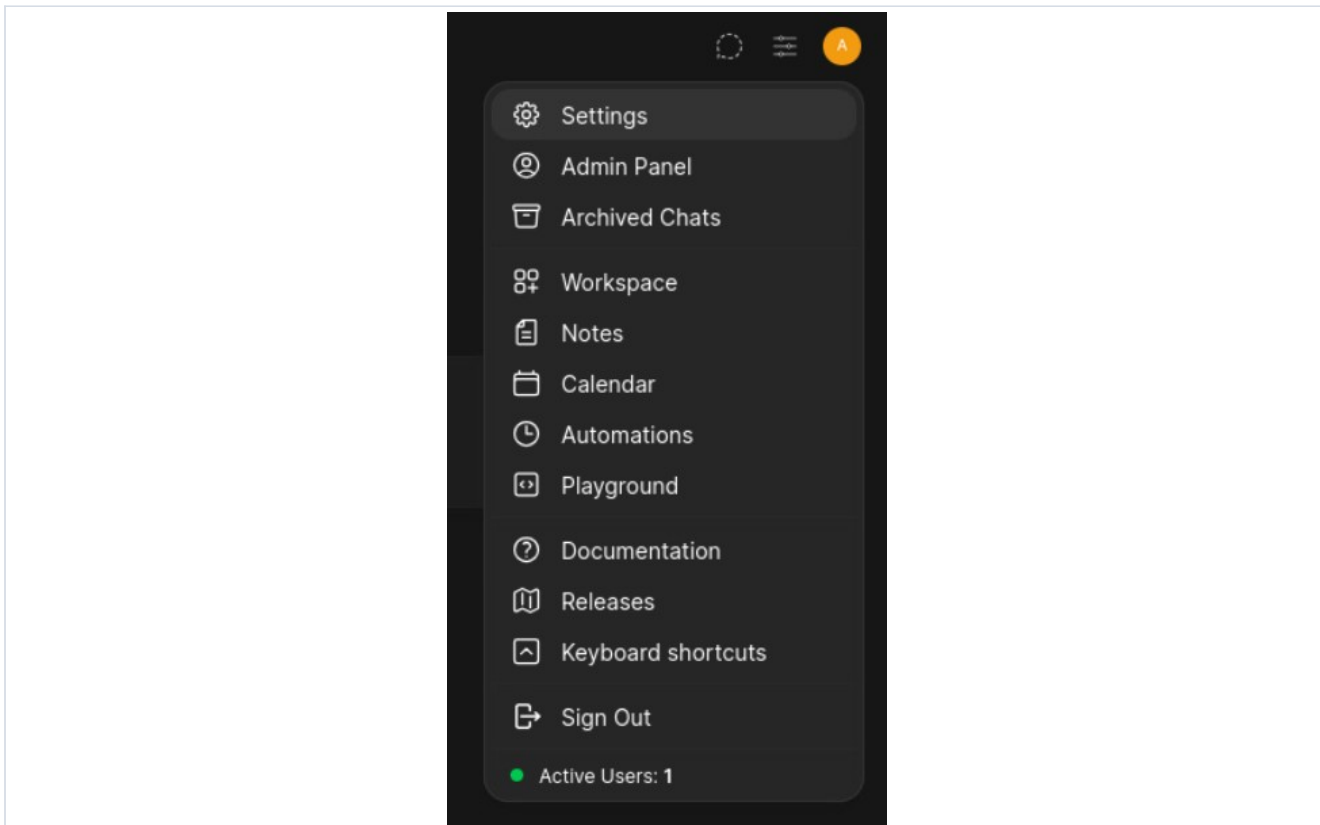


After (on):

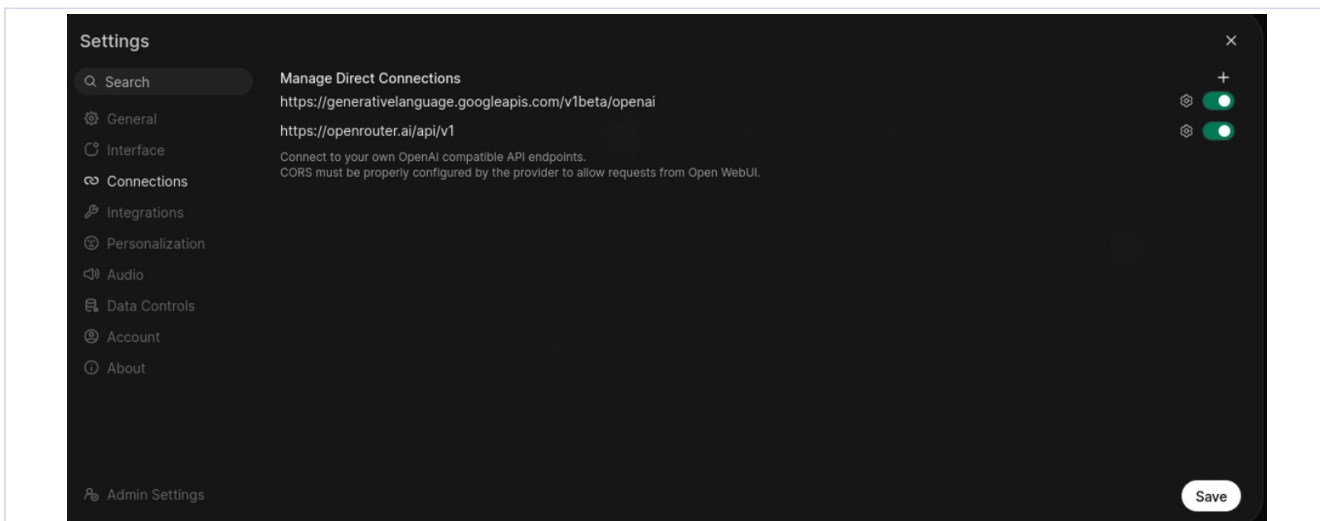


Part C — Add your Groq key to Open WebUI

Step 9 Click your **profile icon** again and choose **Settings**.



Step 10 Click **Connections** in the left menu. Under **Manage Direct Connections**, click the + (plus) button on the right.



Step 11 The **Add Connection** box opens. Fill it in exactly like this, then click **Save**:

- URL: `https://api.groq.com/openai/v1`
- Auth: leave as Bearer, and paste your Groq key (the `gsk_...` one) in the box next to it.
- Model IDs: leave this empty — Open WebUI will then load all Groq models automatically.

ti

Add Connection

×

URL

https://api.groq.com/openai/v1

↺ ☒

Auth

Bearer

▼

gsk_pDpSMXaOghY4r9isb9D7WGdyb3FYPAXRPzc'

🗑

Prefix ID

Prefix ID

Model IDs

Leave empty to include all models from "https://api.groq.com/openai/v1/models" endpoint

Add a model ID

+

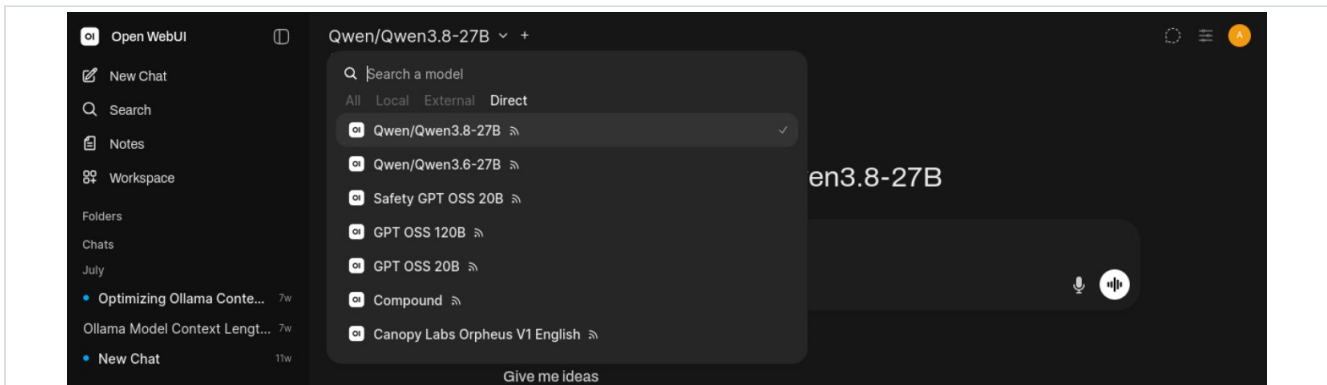
Tags

Add a tag...

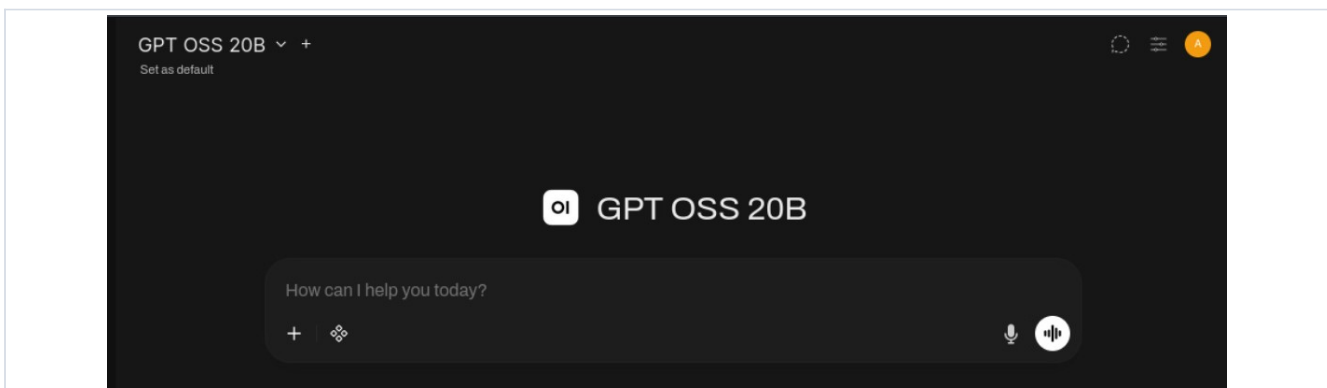
Save

Part D — Select GPT OSS 20B and start

Step 12 Go back to the chat screen and click the **model name** at the top. In the dropdown, click the **Direct** tab to see your Groq models.



Step 13 Select **GPT OSS 20B** from the list. That's the model to use for the practicals.

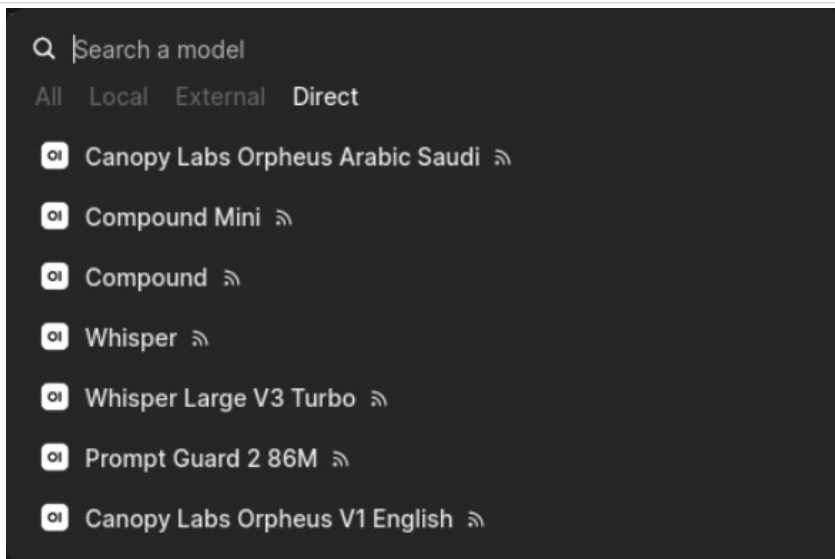


Tip: click "Set as default" (under the model name) so GPT OSS 20B is already chosen every time you open a new chat.

Which models to use and which to skip

Your Groq connection loads several models. Only a couple are meant for these practicals — the rest are for audio or safety and will not answer normal prompts.

- **USE** — GPT OSS 20B: your main model for every practical (fast, light on the free limits).
- **BACKUP** — GPT OSS 120B: switch to this only if you want longer, more detailed answers.
- **SKIP** — Whisper / Whisper Large V3 Turbo: these are speech-to-text (audio), not chat.
- **SKIP** — Canopy Labs Orpheus (English / Arabic): these are text-to-speech (voice), not chat.
- **SKIP** — Prompt Guard / Safety GPT OSS: these are security filters, not chat models.



If you hit a limit

The free Groq plan allows plenty for a day of practicals, but it has a per-minute limit. If you see a rate-limit message:

- Wait about a minute and send again.
- For the big document (RAG) steps, make sure you are on GPT OSS 20B, not the 120B model.
- Don't fire several large prompts within the same minute — space them out slightly.

You're all set — GPT OSS 20B is connected and ready for the practicals.